

Marcus Finlay

Mechatronics Engineer

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EDUCATION

McMaster University

Bachelor of Engineering, Co-op

Cumulative GPA: 3.8 /4.0

Expected Graduation: April 2029

- **Awards:** Deans' Honor List, The Faculty of Engineering Achievement Award
- **Relevant Coursework:** Engineering Design, Graphical Design, Computer Programming, Material Science

WORK EXPERIENCE

Stoney Creek Furniture

Warehouse Assistant

Jun 2022 – Aug 2024

Hamilton, ON

- Collaborated with team members to optimize customer pick-up process, resulting in an increase in customer satisfaction ratings.
- Managed the assembly and disassembly of furniture for customer pick-up, ensuring quality and adherence to company standards.
- Achieved certification on a Class 2 Forklift to efficiently retrieve furniture from high storage racks.
- Implemented waste management procedures by using various compactors and decreased overall warehouse waste production.

PROJECTS

[Airport Logistics](#) | *Autodesk Inventor, Python, 3D Printing*

- Designed a system to facilitate transportation and sorting of airport luggage, using core engineering design principles
- Developed a linear actuator-powered sliding mechanism to efficiently transfer luggage from the airport baggage tunnel to the plane storage area.
- Implemented data processing functions to analyze passenger and flight records, identifying oversold tickets, baggage violations, and delays, while generating visual summaries using Turtle graphics.
- Programed a Q-arm robotic system to automate luggage sorting by scanning barcodes, identifying compliance, and placing baggage in designated locations.
- Collaborated with team members to define design requirements and ensure timely delivery of key project milestones

[Hospital Wastewater Filtration](#) | *Ansys Granta EduPack*

- Designed a filtration system to remove ciprofloxacin from hospital wastewater, addressing environmental and health concerns linked to antibiotic-resistant bacteria.
- Conducted material selection using the Ashby method in Ansys Granta EduPack, optimizing for strength, cost-efficiency, and environmental impact.
- Performed porosity calculations to balance filtration efficiency and structural integrity, ensuring compliance with treatment facility requirements.
- Utilized Eco Audit tools to evaluate the carbon footprint of the filter design, aligning with sustainability goals.
- Collaborated with a team to develop technical objectives, create an objective tree, and deliver comprehensive design documentation and a final presentation.

ACTIVITIES

[MDL Designathon 2025](#) | *Autodesk Inventor*

- Conceptualized and designed an attachment for a Roomba-like device using Autodesk Inventor, enabling efficient dusting of baseboards and walls.
- Presented the design process and functionality of the attachment to a panel of judges, effectively explaining its design and usage.

SKILLS

Programming: Python, HTML, CSS, Arduino

Design & Modeling: MATLAB, Autodesk Inventor, Prusa Slicer, Ansys Granta EduPack

Office Tools: Microsoft Word, Microsoft PowerPoint, Microsoft Excel